

Mapping Serotonergic Dynamics using Drug-Modulated Molecular Connectivity

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Tudor M Ionescu, Mario Amend, Rabikul Hafiz, Andreas Maurer, Bharat Biswal Biswal, Hans F Wehrl, Kristina Herfert ✉

Werner Siemens Imaging Center, Department of Preclinical Imaging and Radiopharmacy, Eberhard Karls University Tuebingen, Germany • Department of Biomedical Engineering, New Jersey Institute of Technology, University Heights, Newark, New Jersey, USA

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Abstract

Understanding the complex workings of the brain is one of the most significant challenges in neuroscience, providing insights into the healthy brain, diseases, and the effects of potential therapeutics. A major challenge in this field is the limitations of traditional brain imaging techniques, which often deliver only a part of the complex puzzle of brain function. Our research employs a novel approach named “Molecular Connectivity” (MC), which merges the strengths of various imaging methods to offer a comprehensive view of how molecules interact within the brain and affect its function.

This innovative technique bridges the gap between functional magnetic resonance imaging (fMRI), known for its ability to monitor brain activity by tracking blood flow, and positron emission tomography (PET), which can depict specific molecular changes. By integrating these methods, we can better understand the far-reaching impacts of drugs on the brain. Our study focuses on the application of dynamic [^{11}C]DASB PET scans to map the distribution of serotonin transporters, a key player in regulating mood and emotions, and examines how these are altered following the use of methylenedioxymethamphetamine (MDMA), commonly known as ecstasy.

Through a detailed analysis comparing MC with traditional measures of brain connectivity, we uncover significant patterns that closely align with physiological changes. Our results reveal clear changes in molecular connectivity after a single dose of MDMA, establishing a direct link between the drug's effects on serotonin transporter occupancy and changes in the brain's functional network.

This work not only offers a novel methodology for the in-depth study of brain function at the molecular level but also opens new pathways for understanding how drugs modulate brain activity.

eLife assessment

This **important** paper on measuring molecular connectivity using combined serotonin PET and resting-state fMRI provides both novel methods for studying the brain as well as insights into the effects of ecstasy administration. The methods are **solid**, with a few doubts that need to be dispelled surrounding the high anaesthetic dose used.

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Introduction

The field of neuroscience has witnessed remarkable advancements over the past decade, particularly propelled by the advent of innovative imaging techniques. The simultaneous application of positron emission tomography (PET) and magnetic resonance imaging (MRI) has revolutionized our ability to concurrently assess brain function across various physiological dimensions [1, 2]. This is especially true in the context of drug effect evaluations. The combination of PET with functional MRI (fMRI) has opened a plethora of investigative avenues. Techniques such as resting-state functional connectivity (rs-FC) [3] or pharmacological fMRI (phMRI) [4] are now complemented by the capability of PET to quantify molecular changes, including alterations in receptor and transporter availability [5].

While fMRI provides high spatial and temporal resolution, the interpretation of its readout necessitates caution. The Blood-Oxygen-Level-Dependent (BOLD) signal used in fMRI indirectly captures neuronal changes through neurovascular coupling [6], online reveals the hemodynamic consequences of molecular-level drug effects. The integration of simultaneous PET acquisition can bridge this interpretative gap by offering essential molecular insights, particularly regarding transporter or receptor alterations. Typically, PET, used either independently or simultaneously to fMRI, has been mainly used in pharmacological studies for illustrating quantitative shifts in neuroreceptor or transporter availabilities [7]. Several studies have also explored the interregional coherence of PET tracer signals [8–10], an approach akin to fMRI-derived rs-FC. While subject-wise metabolic connectivity using [^{18}F]FDG-PET has been established through the temporal correlation of regional PET signals [1, 8], studies employing transporter or receptor tracers have predominantly focused on interregional binding coherences across subjects using static scans [9, 10]. The concept and feasibility of molecular connectivity (MC) through the temporal correlation of dynamic binding potentials of transporter or receptor tracers has yet to be explored.

In our study, we investigate the feasibility of deriving MC from dynamic [^{11}C]DASB-PET scans acquired simultaneously with fMRI in rats. We divided the rats into two cohorts: a baseline group and a pharmacological application group, exposed to 3,4-methyldioxymetamphetamine (MDMA). The baseline cohort served to assess the feasibility of the novel methodology, contrasting it with traditional fMRI-derived rs-FC with a specific focus on temporal stability. We postulated that dynamic [^{11}C]DASB PET temporal fluctuations could be harnessed for connectivity data, in a manner similar to hemodynamic rs-FC, using seed-based and independent component analysis (ICA).

The second cohort, subjected to an MDMA challenge allowed us to evaluate the utility of our novel approach. We aimed to outline the effects of MDMA by integrating the innovative MC concept with established analysis techniques. Our primary objective of this research was to elucidate the

potential of PET-derived MC in conjunction with simultaneous PET/fMRI, exploring the avenues this methodology could open for future diagnostic and drug development studies.

Material and Methods

Our study reevaluates two distinct datasets previously published by our group [11, 12], to simultaneously explore FC and MC both at baseline conditions [12] and following MDMA administration [11]. For detailed descriptions on animal handling, experimental setups, and data acquisition procedures, please refer to these earlier publications.

Animals

A total of 41 male Lewis rats were obtained from Charles River Laboratories (Sulzfeld, Germany). Thirty rats underwent baseline [¹¹C]DASB PET/fMRI scans without any pharmacological intervention, while 11 rats underwent [¹¹C]DASB PET/fMRI scans including an acute MDMA challenge. All experiments were conducted in compliance with German federal regulations for experimental animals and received approval from the Regierungspräsidium Tübingen.

Simultaneous PET/fMRI Experiments

The rats were subjected to simultaneous PET/fMRI experiments involving 1.3% isoflurane anesthesia, tail vein catheterization, positioning on a temperature-controlled bed, and monitoring vital signs. The scans were performed using a 7T small-animal ClinScan MRI scanner (Bruker Biospin, Ettlingen, Germany) with a custom-developed PET insert [13]. Both the scanning protocol and the sequence parameters have been outlined in detail in our previous publication [11]. The MDMA cohort received a pharmacological MDMA challenge of 3.2 mg/kg intravenously 40 minutes after tracer injection.

Data Preprocessing and Analysis

Data preprocessing followed established protocols, including steps such as realignment, mask creation, coregistration, spatial normalization, signal cleaning, and spatial smoothing, as detailed in our previous work [11]. For the MDMA dataset, PET scans were analyzed for early and late effects post-challenge using the general linear model (GLM) available in SPM. For both datasets the baseline was defined 30 to 40 minutes after scan start. For the fMRI data a first level analysis was applied to the individual scans without a high-pass filter (the filter was set to “Inf”). Statistical parametric maps were generated post GLM parameter estimation using contrast vectors. Group-level analysis involved a one-sample t-test on the subject-level statistical parametric maps ($p < 0.05$, one-sided, family wise error / FWE-adjusted).

Static PET scans were generated by summing dynamic frames over defined periods for 10-minute periods after the MDMA challenge (50-60 minutes to investigate early effects, and 70-80 minutes to investigate late MDMA effects). Two-sample t-maps were calculated between the normalized [¹¹C]DASB uptakes of (1) the baseline scan period and the early effect time period and of (2) the early effect time period and the late effect time period ($p < 0.05$, FWE-adjusted).

All group-level t-maps underwent voxel-wise signal quantification to determine the regional contributions of 48 regions selected according to the Schiffer atlas [14]. Average t-scores and standard deviations of all voxels were calculated.

Functional Connectivity Analysis

Functional connectivity was determined using a seed-based analysis approach. The mean time series of the pre-processed BOLD-fMRI signals for each dataset across all regions (refer to *Supplementary Table 1* for the list of regions) were extracted using SPM toolbox Marseille Boîte À Région d'Intérêt (MarsBaR). Pairwise Pearson's correlation coefficients were calculated between each pair of mean regional time-series for every dataset. The Pearson's r coefficients were converted into z -values using Fischer's transformation for group-level analysis. The Fischer's z -transformed correlation coefficients were then used to generate mean correlation matrices for both cohorts [15].

Molecular connectivity analysis

The mean [^{11}C]DASB signal from the preprocessed PET datasets was extracted from the designed regions, including the 48 regions used for fMRI data analysis and the cerebellum using MarsBaR. Binding potentials were calculated frame-wise for all dynamic PET scans using the DVR-1 (equation 1) to generate regional BP_{ND} values using the cerebellum as a reference region [16]:

$$\text{BP}_{\text{ND}} = \frac{V_T - V_{\text{ND}}}{V_{\text{ND}}} = \frac{V_T}{V_{\text{ND}}} - 1 = \text{DVR} - 1,$$

where:

- BP_{ND} is the binding potential
- V_T is the total volume of distribution
- V_{ND} is the volume of distribution in a reference tissue
- DVR is the relative volume of distribution

To calculate MC, we discarded the first 20 minutes of every scan, which were dominated by perfusion effects and applied a detrending approach on the remaining 60 minutes in order to obtain temporally stable values (for further details, please refer to Supplementary Methods and Supplementary Figure 1). The BP_{ND} time courses were then used to calculate MC as described above for fMRI: subject-level correlation matrices between all regional time courses were generated and z -transformed correlation coefficients were used to calculate mean correlation matrices.

Independent Component Analysis

Group ICA (GIFT toolbox, MIALAB, University of New Mexico, Albuquerque, NM, USA) was used for ICA of the baseline group. Both the fMRI and PET preprocessed data sets were investigated between 30 and 80 minutes after start of data acquisition. For fMRI, we selected 10 independent components, while we started with two components for PET and increased the number to ten components to thoroughly dissect the varying components within the signal. Components were thresholded at a z -value ≥ 1.96 ($p\text{-value} \leq 0.05$) [17] and average z -scores and standard deviations were calculated for each component. These components' physiological significance was further explored by contrasting them with the regional [^{11}C]DASB changes induced by MDMA. Accordingly, the z -scores of the independent components generated in the baseline cohort were correlated with the early and late regional [^{11}C]DASB changes induced by MDMA, measured using t -scores.

Results

Comparability of MC and FC in spatial contexts and over time

We first aimed to evaluate whether MC aligns spatially with FC, possesses similar graph theory properties and provides consistent temporal readouts throughout the scan duration in the baseline group (**Figure 1** [↗](#)).

We found moderate, but significant correlation between the edge-level MC and FC ($r = 0.51$, $p < 0.001$, **Figure 1A and B** [↗](#)). Furthermore, both connectomes revealed small-world properties at group-level, with coefficients higher than 1 (**Figure 1C** [↗](#)). At subject level, three molecular and four functional connectomes fell below the threshold of 1 for the small-world coefficient. A significant consistency was observed between early and late scan-derived connectomes (**Figure 1D** [↗](#) for FC, G for MC), with FC having a slight edge (**Figure 1E** [↗](#), $r = 0.96$) over MC (**Figure 1H** [↗](#), $r = 0.8$). While both, FC and MC maintained steady correlation intensities, there was a negligible decline over the scan duration (**Figure 1F** [↗](#) and **1I** [↗](#)).

Deciphering spatial characteristics of FC and MC using ICA

After establishing the feasibility of obtaining temporally stable readouts using the ROI-to-ROI approach, we employed a data-driven approach, using ICA, to compare spatial characteristics of FC and MC (**Figure 2** [↗](#)).

We extracted ten group ICs from the fMRI data (**Figure 2A** [↗](#)), revealing known canonical resting-state networks, such as the posterior default-mode-like network (IC1-5, red), sensorimotor networks (IC1-5, green and purple), the anterior default-mode-like network (IC6-10, yellow) or the visual network (IC6-10, red). With unpredictability for the number of ICs suitable for MC IC extraction, we started with two components (**Figure 2B** [↗](#)). IC1 (orange) comprised both subcortical and cortical anterior brain regions, including the nucleus accumbens, amygdala, cingulate cortex, caudate putamen, orbitofrontal cortex and medial prefrontal cortex, while IC2 (blue) primarily received contributions from deeper posterior areas, such as the midbrain, thalamus, hypothalamus, periaqueductal gray and medulla. Interestingly, when we extracted 10 independent components, to mimic the number of components used for the FC data, we found that the initial anterior component split in four different ICs and the initial posterior IC in three different ICs. A relatively clear spatial segregation can be seen for the newly-formed ICs, for instance the three posterior components being extracted from specific regions (green mainly from medulla, red from hypothalamus and part of the midbrain, blue from midbrain and thalamus).

MDMA-induced changes of ICA-derived molecular connectivity

Next, we aimed to explore the relationship between molecular changes in SERT availability and the molecular connectome derived from ICA, induced by an acute MDMA administration (**Figure 3** [↗](#)).

Two IC extracted from the MC showed good overlap with regions associated with the salience network (IC1) and with those having high SERT densities (IC2). Therefore, we defined the regions contributing strongly to IC1 as salience network (CPu, Cg, NAc, Amyg, Ins, mPFC) and those with strong signals in IC2 as SERT network (VTA, Th, MB, PAG, Hypo). Interestingly, MDMA induced immediate strong decreases in all SERT network regions, salience areas exhibited a delay by approximately 10 minutes (**Figure 3A** [↗](#)). Voxel-level analysis showed clear spatial overlaps between early MDMA responsive regions and those from the posterior IC, with delayed regions mirroring the anterior IC reminiscent of the salience network (**Figure 3B** [↗](#)). To quantify the striking spatial similarity between the baseline independent components and the spatiotemporal characteristics of [^{11}C]DASB changes after MDMA exposure, we show highly significant correlations between the z-scores of the posterior and anterior ICs and the t-scores of late and early MDMA effects on [^{11}C]DASB alterations, respectively ($p < 0.0001$, **Figure 3C** [↗](#)).

Connectome comparison

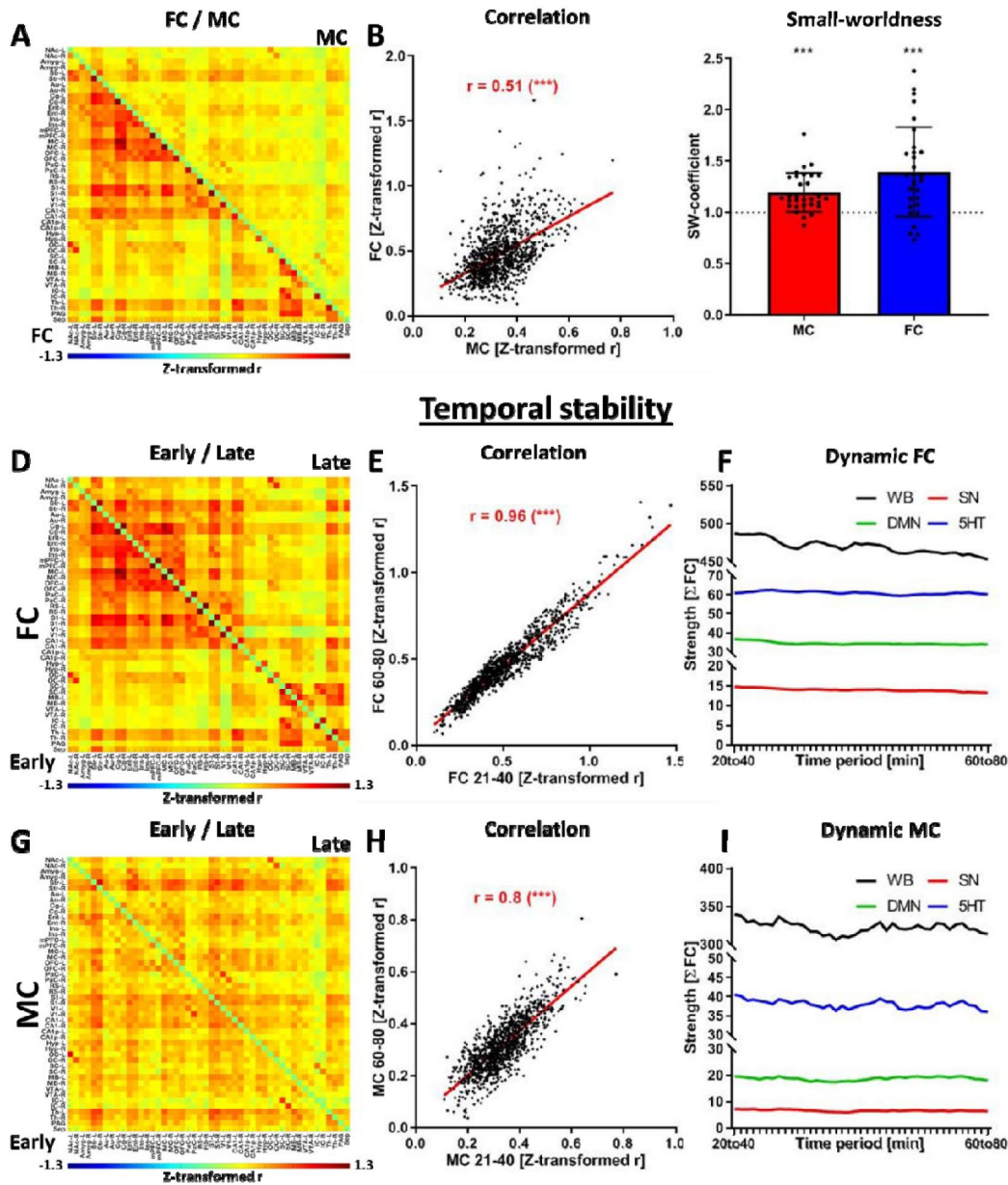


Figure 1

Evaluation of seed-based MC.

(A) Correlation matrix indicating whole-brain FC (beneath diagonal) and MC (above diagonal). Correlations not surviving significance testing with multiple comparison corrections were set to zero ($p < 0.05$, FWE correction). (B) Scatter plot and correlation between MC and FC edges. (C) Small-world coefficients for all subjects and group-level one-sample t-test against the value of 1 ($SW > 1$ is considered as indicative of small-world properties). Comparison of (D) FC and (G) MC early (20-40 minutes after scan start, below diagonal) and late (60-80 minutes after scan start, above diagonal). The similarities of early and late readouts were quantified for both (E) FC and (H) MC. Temporal stability of both (F) FC and (I) MC was evaluated using a sliding window approach including 20-minute windows between 20 and 80 minutes after scan start. Abbreviations: FC = fMRI-derived hemodynamic functional connectivity, MC = [^{11}C]DASB PET-derived molecular connectivity.

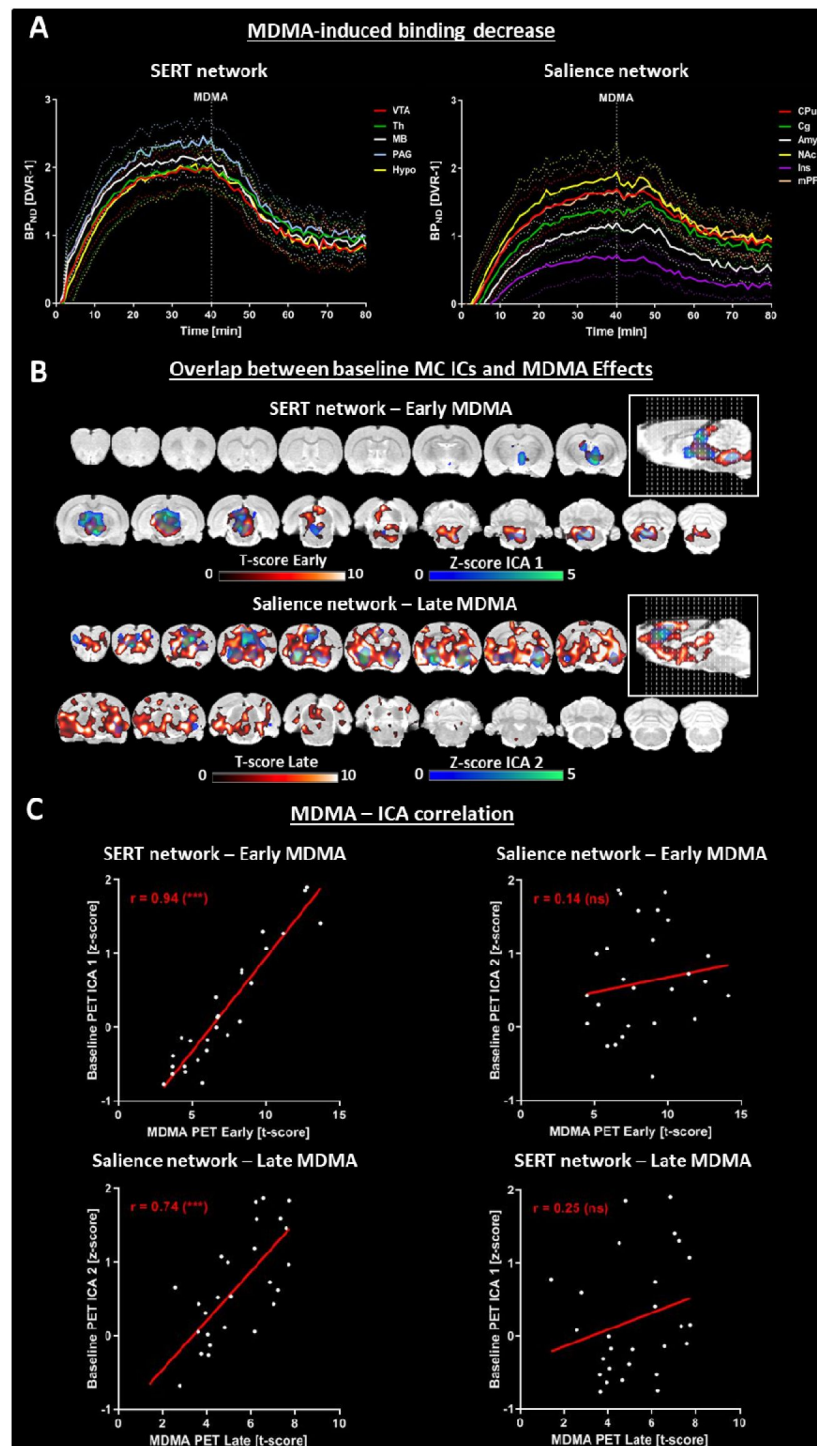


Figure 3

Comparison of MDMA-induced [¹¹C]DASB alterations.

(A) Left panel: Dynamic binding potentials of regions comprising SERT network, defined by IC 1 in validation cohort. Right panel: Dynamic binding potentials of regions comprising salience network, defined by IC 1 in validation cohort. (B) Overlap between independent components extracted from the validation cohort (IC 1 = SERT network; IC 2 = salience network) and the early and late effects of MDMA respectively. (C) Pair-wise correlations between regional z-scores of the ICs extracted from validation cohort and regional t-scores of early and late MDMA effects. (***) indicates $p < 0.001$, ns = non-significant). Abbreviations: SERT = serotonin transporter, ICA = independent component analysis; for abbreviations of regions please refer to *Supplementary Information*.

Finally, we investigated relationships between SERT availability changes and MC reductions following the acute MDMA challenge (**Figure 4** [↗](#)).

We found that the decreases in MC and in [^{11}C]DASB BP_{ND} following MDMA application (**Figure 4A** [↗](#)) showed only low correlation ($r = 0.29$, **Figure 4B** [↗](#)). While decreases in SERT availability exhibited a strong anterior-posterior gradient, being most pronounced in areas with high SERT availability, such as the MB, VTA, Pons or PAG, MC encompassed regions across the brain to similar extents. Specifically, while the significance of [^{11}C]DASB reductions in the Ins was very low across investigated regions, the Ins showed highest effects among regions for MC, while strong SERT occupancy effects in the IC and SC did not translate in very prominent reductions in the respective global MC of the two regions.

MDMA-induced changes of seed based molecular connectivity

Next, we performed a seed-based analysis and aimed to compare changes in FC and MC after an MDMA pharmacological challenge on the salience network and on regions with high SERT binding (**Figure 5** [↗](#)).

We observed a small decrease in the SN for FC (16% at the end of the scan) and almost constant FC of the SERT network (<5% decrease) following MDMA, as shown in **Figure 5A** [↗](#) at edge and node level and **Figure 5C** [↗](#) at network level ($p > 0.05$). For MC, we observed profound reductions in the SERT network (**Figure 5C and D** [↗](#)), emphasizing an acute and spatially specific effect of MDMA on MC.

Discussion

The mammalian brain operates on diverse physiological, spatial and temporal scales. FC via BOLD-fMRI offers insights into coherent functional brain networks, but its complexity and indirect link to neural activity highlight the need for more direct methodologies. In this context, the concept of MC using [^{11}C]DASB PET, as introduced in our study, provides a more direct and complementary perspective on brain organization and its response to external stimuli, such as MDMA.

Physiological basis

Our findings suggest that [^{11}C]DASB binding reflect the interplay of serotonin levels and SERT dynamics [[18](#) [↗](#)]. Supporting the competition model, evidence indicates that endogenous serotonin competes with tracers for binding sites, affecting tracer binding [[19](#) [↗](#)]. However, contrasting results in various studies highlight the complexity of this interaction [[20](#) [↗](#)] [[21](#) [↗](#)] [[22](#) [↗](#)]. The internalization model suggests serotonin levels influence SERT internalization, impacting [^{11}C]DASB binding [[18](#) [↗](#), [23](#) [↗](#), [24](#) [↗](#)]. While supporting evidence exists, further exploration is needed to fully understand these dynamics, especially during resting-state [[25](#) [↗](#)]. Some models on this aspect have proposed a regulatory function of the raphe nuclei in maintaining serotonin fluctuations over several temporal scales at rest [[26](#) [↗](#)]. Remarkably, fast microdialysis has resolved multiple spontaneous surges of up to 1500-fold of the basal serotonin occurring during 30-minute intervals [[27](#) [↗](#)]. Additionally, the same study has indicated that SERT expression is essential for the spontaneous surges, reduced SERT drastically decreasing serotonin spiking. Thus, it is feasible that the correlated temporal fluctuations captured by dynamic [^{11}C]DASB at least partly reflect the role of the serotonergic system in the resting activity of the brain.

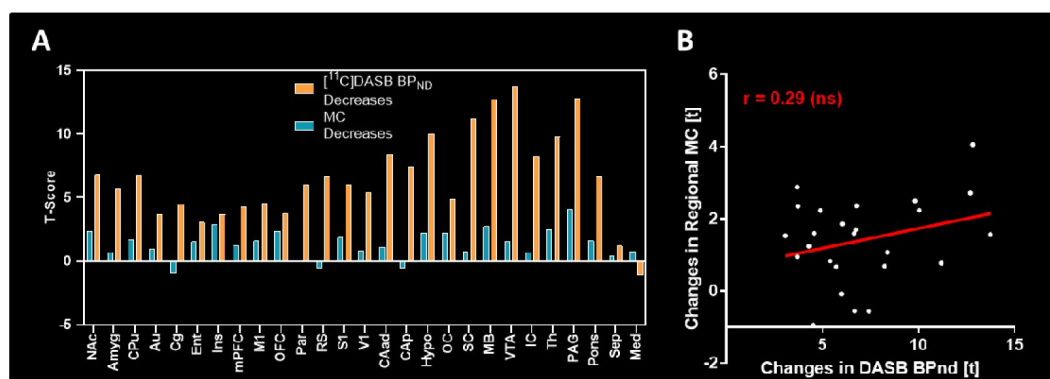


Figure 4

Comparison of MC and BP_{ND} changes.

A Reductions in BP_{ND} following MDMA (orange) and MC strength (blue) were compared. **B** T-scores derived from each approach show only low correlations.

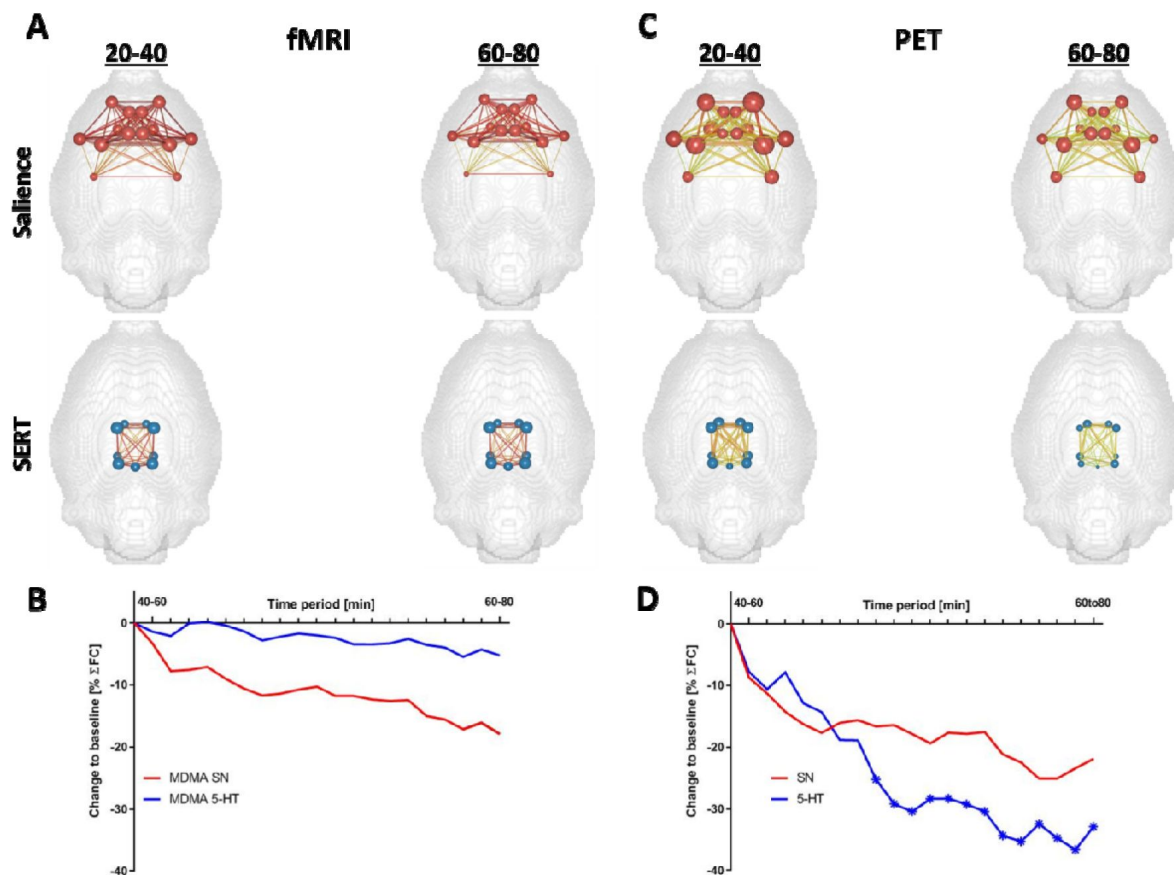


Figure 5

MDMA effects on seed-based FC and MC of the salience and SERT networks.

(A) FC and (C) MC brain networks depicting edge and node strengths of the salience network and SERT network at baseline (20-40 min after scan start) and after MDMA (60-80 min after scan start). (B) FC and (D) MC time-resolved salience and SERT network strengths computed using sliding windows. Asterisks indicate significant ($p < 0.05$, FDR-corrected) changes to baseline (time-point zero, corresponding to 20-40 minutes after scan start). Abbreviations: SN = salience network, SERT = SERT network.

Implications for whole-brain serotonergic system

Our graph-theory analysis revealed comparable whole-brain organization between hemodynamic and serotonergic connectomes, although we only found moderate correlation between the edges of the two measurements. Notably, the correlations in [^{11}C]DASB bindings differed from BOLD-fMRI-derived correlations. For BOLD-fMRI, ICA revealed classic RSNs, widely reported for both rats [28] and humans [29], including the default-mode network, the sensory network and the motor network. In contrast, the ICA analysis of PET data revealed two distinct anatomical modules. The first component included primarily the brainstem, parts of the midbrain and the thalamus. The second component comprised regions of the limbic system, such as the striatum, amygdala, insular, cingulate and prefrontal cortices, reflecting the diverse projections of the raphe nuclei. MDMA's interaction with SERT, as elucidated by [^{11}C]DASB, shows region-specific temporal effects, with immediate decreases in binding in the brainstem and posterior subcortical regions, and delayed effects in limbic pathways. This suggests that the serotonergic system's response to drugs can be predicted, to an extent, by the observed independent components.

Relevance to prior research

Our findings align with those by Salvan et al [31] on the integration of molecular maps into fMRI data demonstrating how individual serotonergic receptors contribute to network-level activity. Despite our differing methodologies, the receptor activity patterns they found, may also correspond to the different independent components we found from the [^{11}C]DASB PET data. The dual regression approach reported by Salvan et al. has first been reported in the context of pharmacology to delineate the receptor-specific effects of MDMA in humans [32]. The authors found that MDMA specifically decreased FC in the 5HT $_{1A}$ maps in areas which could be ascribed to the human salience network, such as the insula and a collection of medial cortical regions. While not reaching significance, we observed a reduction in salience FC and MC in our data.

Previous studies evaluating the effects of MDMA on FC indicated relatively sparse effects on FC, as confirmed by the readout of this study [33, 34]. Recently, a novel approach [32] has revealed decreased connectivity induced by MDMA in areas associated with SERT and 5-HT $_{1A}$ availability. The increased activity indicated in limbic and cortical structures when controlling for vascular effects is accompanied with decreased salience connectivity, indicating that, while neurons become more active through the drug, they do so in an incoherent manner, which would be in line with the hyperactive, yet abnormal behavior reported for MDMA abuse. Importantly, the potent vascular effects which play a role in modulating the amplitude of the BOLD-fMRI signal may also influence FC readouts, although the extent of such effects are difficult to estimate [11].

Over the past decade, PET studies using [^{11}C]DASB have focused on the associations of serotonin transporter (SERT) availabilities across different brain regions, revealing altered interregional SERT connections in patient cohorts, post-treatment changes, and predictive capacity for treatment response [9, 10, 35] [36] [37]. Our study builds upon this foundation, uncovering significant within-subject temporal associations in [^{11}C]DASB binding, and demonstrates more pronounced alterations in MC compared FC changes induced by drugs, thus highlighting the enhanced utility of our method.

Study limitations and future directions

While our study advances the understanding of MC, the use of anesthetized animal is a limitation, potentially affecting PET and fMRI readouts and their interaction with MDMA. Future studies should involve awake animals to validate our method. Additionally, our multimodal imaging approach, though powerful, cannot fully decipher the mechanisms of interregional coherences in

PET timecourses. Future studies employing sensitive methods to measure neurotransmitter release, could provide deeper insights into the molecular processes underpinning our observations.

Nonetheless, we provide several strong arguments to consider such analyses as presented here in future studies. First, we show that at rest the data are reliable, temporally stable and exhibiting similar graph theory metrics to traditionally calculated functional connectomes. Second, in spite of comparable network-level organizational properties, already at rest MC is only moderately correlated to FC, indicating the complementary nature of both readouts. Third, resting-state MC ICs correlate well with SERT occupancy changes induced by the MDMA challenge. Finally, we show that the changes MDMA elicits on MC are complementary to standardly-calculated [^{11}C]DASB BP_{ND} alterations. Our data indicate that, while changes in BP_{ND} are more pronounced in regions with higher baseline SERT availabilities, MC reveals a more globally distributed measure of tracking serotonergic changes, since also regions, such as the insula, with relatively low SERT expressions, were strongly affected.

Conclusion

This study significantly contributes to integrating molecular data into connectomic frameworks, demonstrating that subject-level MC is reliable and complementary to FC in both resting and pharmacologically challenged states. Our research lays a strong foundation for future investigations into the value and generalizability of PET-derived MC, particularly for understanding drug-induced brain-wide molecular network changes.

Statements & Declarations

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Competing Interests

The authors declare no conflict of interest.

CRediT authorship contribution statement

Tudor M. Ionescu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Mario Amend:** Investigation, Writing – review & editing. **Rakibul Hafiz:** Methodology, Software, Writing – review & editing. **Bharat B. Biswal:** Conceptualization, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Andreas Maurer:** Writing – review & editing. **Hans F. Wehrl:** Conceptualization, Methodology, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Kristina Herfert:** Writing – review & editing, Visualization, Supervision, Project administration, Funding.

Data availability

The data will be made available upon reasonable request.

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Reviewing Editor

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University of Oxford, Oxford, United Kingdom

Senior Editor

Michael Frank

Brown University, Providence, United States of America

Reviewer #1 (Public Review):

This paper by Ionescu et al. applies novel brain connectivity measures based on fMRI and serotonin PET both at baseline and following ecstasy use in rats. There are multiple strengths to this manuscript. First, the use of connectivity measures using temporal correlations of 11C-DASB PET, especially when combined with resting state fMRI, is highly novel and powerful. The effects of ecstasy on molecular connectivity of the serotonin network and salience network are also quite intriguing.

I would like the authors to discuss and justify their use of high-dose (1.3%) isoflurane. A recent consensus paper on rat fMRI (Grandjean et al., "A Consensus Protocol for Functional Connectivity Analysis in the Rat Brain.") found that medetomidine combined with low dose isoflurane provided optimal control of physiology and fMRI signal. To overcome any doubts about the effects of the high-dose anaesthetic I'd encourage the authors to show the results of their functional connectivity specificity using the same or similar image processing protocol as described in that consensus paper. This is especially true since the fMRI ICs in Figure 2A appear fairly restricted.

I'd also be interested to read more about why the cerebellum was chosen as a reference region, given that serotonin is highly expressed in the cerebellum, and what effects the choice

of reference region has on their quantification.

The PET ICs appear less bilateral than the fMRI ICs. Is that simply a thresholding artefact or is it a real signal?

"The data will be made available upon reasonable request" is not sufficient - please deposit the data in an open repository and link to its location.

<https://doi.org/10.7554/eLife.97864.1.sa1>

Reviewer #2 (Public Review):

Summary:

The article aims to describe a novel methodology for the study of brain organization, in comparison to fMRI functional connectivity, under rest vs. controlled pharmacological stimulation.

Strengths:

Solid study design with pharmacological stimulation applied to assess the biological significance of functional and (novel) molecular connectivity estimates.

Provides relevant information on the multivariate organization of serotonergic system in the brain.

Provides relevant information on the sensitivity of traditional (univariate PET analysis, fMRI functional connectivity) and novel (molecular connectivity) methods in measuring pharmacological effects on brain function.

Weaknesses:

While the study protocol is referenced in the paper, it would be useful to at least report whether the study uses bolus, constant infusion, or a combination of the two and the duration of the frames chosen for reconstruction. Minimal details on anesthesia should also be reported, clarifying whether an interaction between the pharmacological agent for anesthesia and MDMA can be expected (whole-brain or in specific regions).

Some terminology is used in a bit unclear way. E.g. "seed-based" usually refers to seed-to-voxel and not ROI-to-ROI analysis, or e.g. it is a bit confusing to have IC1 called SERT network when in fact all ICs derived from DASB data are SERT networks. Perhaps a different wording could be used (IC1 = SERT xxxxx network; IC2= SERT salience network) .

The limited sample size for the rats undergoing pharmacological stimulation which might make the study (potentially) not particularly powerful. This could not be a problem if the MDMA effect observed is particularly consistent across rats. Information on inter-individual variability of FC, MC, and BPND could be provided in this regard.

<https://doi.org/10.7554/eLife.97864.1.sa0>

Author Response:

We thank the reviewers for their insightful feedback. In our revised version of the manuscript, we will address all points raised.

Regarding the preprocessing (Reviewer 1), we agree that the StandardRat pipeline is optimal for newly acquired datasets. However, since this study involves reanalyzing an already

published dataset (Ionescu et al., JNM, 2023), which was preprocessed, analyzed, and published before the StandardRat paper, we aimed to maintain the same preprocessing. This approach allows for consistent interpretation of the readout regarding functional and molecular connectivity in the context of our previously published findings. Nonetheless, we agree that providing full access to the data will enable other researchers to reproduce our results using the StandardRat preprocessing pipeline and perform additional analyses on this rich dataset. Therefore, we will provide full access to the data via an open repository, as the reviewer suggested.

Regarding anesthesia, we acknowledge that this is a limitation of our study, as more recent studies have indicated superior protocols. However, we and others have shown that, while not ideal, isoflurane at the used dose maintains stable physiology and does not cause burst suppression in rats. We will amend our discussion to reflect these points.

Regarding the other points, we will amend the manuscript to provide more detail on the experimental design, including the tracer application as suggested by Reviewer 2, and clarify parts of the analysis that are unclear in the current version. Additionally, we agree with Reviewer 2 that our current terminology may cause confusion, and we will amend it accordingly. We will also discuss the other points raised by the reviewers, such as the reduced sample size for the pharmacological cohort as limitations in our discussion.

Thank you for your understanding and the opportunity to improve our manuscript.

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